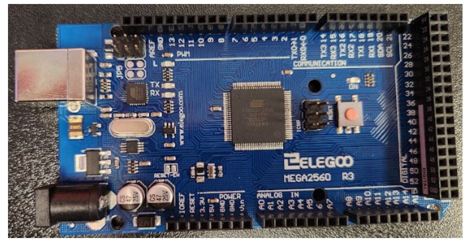
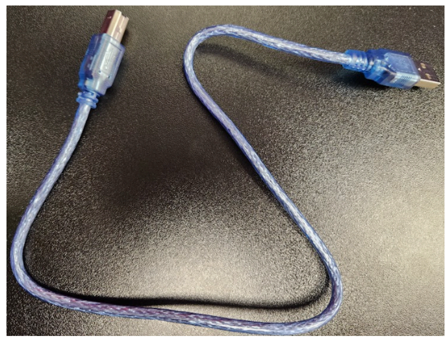
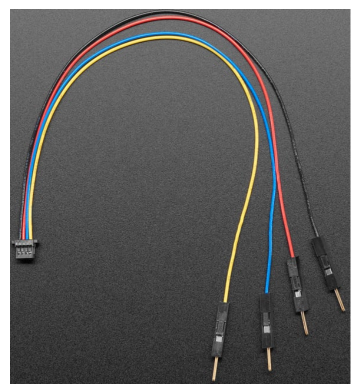
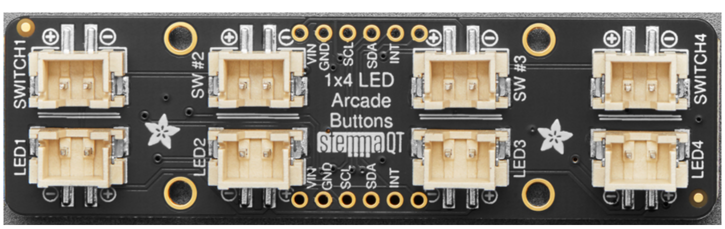
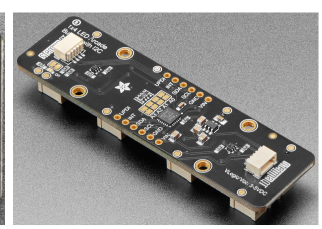
Hello! Welcome to your guide for connecting the buttons on your button board. Don't worry if you've never done any electronics before! We'll go slowly and explain everything clearly.

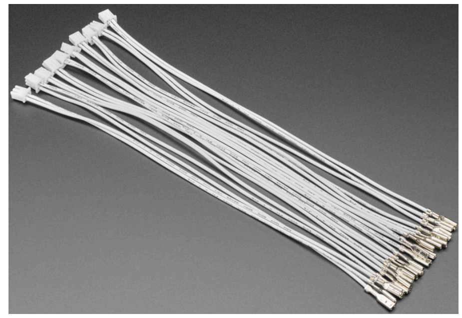
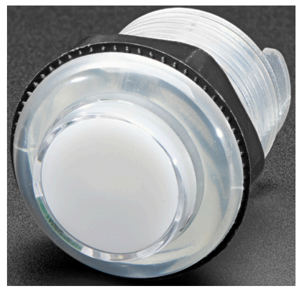
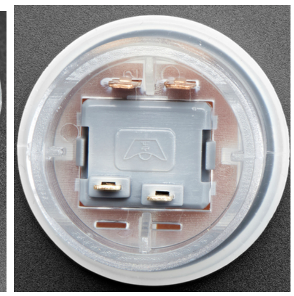
What You'll Need

1. Elegoo Mega2560 R3 microcontroller board 

2. USB cable for the Mega2560 
3. STEMMA QT / Qwiic JST SH 4-pin to Premium Male Headers Cable 
4. Adafruit LED Arcade Button 1x4 - STEMMA QT I2C Breakout - STEMMA QT / Qwiic





5. Arcade Button Quick-Connect Wire Pairs - 0.11" (10 pack) 
6. 4x Arcade Button with LED - 30mm Translucent  

Step 1: Understanding Your Buttons

Look at the back of one of your buttons. You'll notice an image surrounded by two sets of prongs:

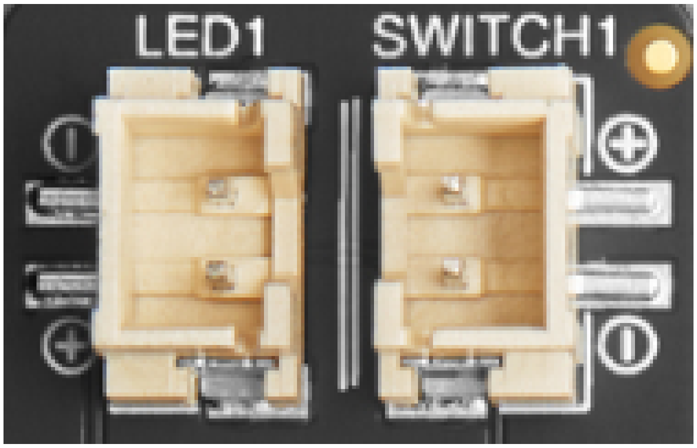
1. Prongs in line with each other near the flat side of the image→ These control the LED inside the button.
2. Prongs that are staggered near the open side of the image→ These control the switch (the part you press).

Each button needs two Quick-Connect wire pairs, so a total of four wires connecting it to the I2C Breakout.

Step 2: Wiring Each Button

1. Take your Quick-Connect wires.
2. Connect the LED wires from the button to the I2C Breakout:
   - Positive (+) wire → prong in line with the other prong (flat side of the image).
   - Negative (–) wire → prong in line (same side).
3. Connect the switch wires:
   - Positive (+) wire → prong staggered (near the open side of the image).
   - Negative (–) wire → prong staggered (same side).
4. Repeat this for all four buttons.

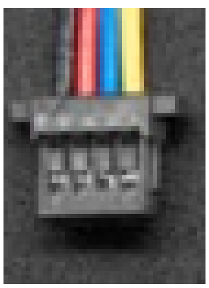
   Tip: Make sure wires are snug on the prongs, but don't force them. If they feel stuck, wiggle gently.



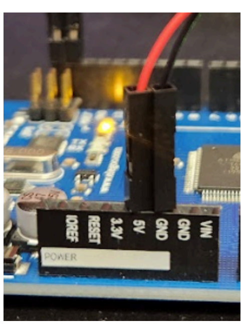
The positive and negative leads from the LED and the Switch correspond to the positive and negative leads on the buttons. In this configuration, the button's negative connections are on the left and the positive connections are on the right.

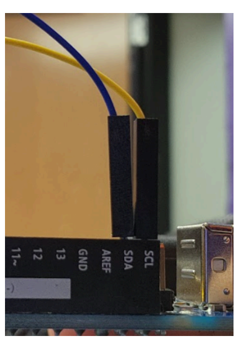
Step 3: Connecting the I2C Breakout

1. Take the STEMMA QT / Qwiic JST SH 4-pin to Premium Male Headers Cable and plug it into the back of the Adafruit LED Arcade Button 1x4 I2C Breakout.

    → 

2. Connect the other ends of the cable to your Mega2560:
   - Red wire → 5V
   - Black wire → GND (ground)
   - Yellow wire → SCL
   - Blue wire → SDA



Step 4: Connecting to Your Computer

1. Plug your USB cable into the Mega2560 and then into your computer.
2. Open the Arduino IDE.
3. Make sure you have installed these libraries:
    - Adafruit seesaw
    - Adafruit BusIO
    - Adafruit ST7735 and ST7789
    - Adafruit GFX

These libraries make it easy for the computer to talk to your buttons and LEDs.

For coding reference:

Buttons	LEDs (PWM)
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Button/Switch 1 -> pin 18	LED 1 -> pin 12
Button/Switch 2 -> pin 19	LED 2 -> pin 13
Button/Switch 3 -> pin 20	LED 3 -> pin 0
Button/Switch 4 -> pin 2	LED 4 -> pin 1